

ajedrezUPV - Project Documentation

Artificial Intelligence Chess Engine Project

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Contents

1 Project Overview

ajedrezUPV is an AI chess project from UPV. Focus: decision algorithms, heuristic evaluation, game engine programming.

Key Features

- Custom chess engine (C++23)
- Alpha-Beta pruning algorithm
- UCI notation support
- Neural network integration
- Modular and extensible code
- Human vs AI gameplay

2 Repository Structure

Reorganized with English names:

Path	Description
.	Root
README.md	Main documentation
LICENSE	CC BY-NC 4.0
docs/	Documentation FEN.pdf PROJECT_DOCUMENTATION.tex PROJECT_DOCUMENTATION.pdf
data/	Data files dataset.csv mnist_loader.py
models/	Models modelo.py
tests/	Tests prova1.py
implementations/	C++ engine alphabetaBot.cpp botConMinimaxOptimizado.cpp NodeMove.cpp/h redNeuronalSquareRook.cpp
traditional_evaluation/	Evaluators EndgameEvaluator.cpp/h Evaluator.cpp/h GeneralEvaluator.cpp/h OpeningEvaluator.cpp/h
neural_network/	Neural Network jsonExtractor.cpp json.hpp modificacionesRedNeuronal.cpp
neural_network_jr/	NN Junior jsonExtractorJR.cpp redNeuronalJR.cpp
connect_four/	Connect Four AI main.ipynb model.py test-case-parser.py test/
connect_three/	Connect Three tres_en_ratlla1.py quatre_en_ratlla.py
lichess_db/	Lichess DB chunk_000 to chunk_019.jsonl

3 Chess Engine

C++ implementation with Alpha-Beta algorithm using Disservin library.

Key Components

- `alphabetaBot.cpp` - Main game loop, UCI input, console display
- `NodeMove.cpp/h` - Game tree node with `alphaBeta()` method
- `GeneralEvaluator.cpp` - Position evaluation

- `OpeningEvaluator.cpp` - Opening heuristics
- `EndgameEvaluator.cpp` - Endgame heuristics

4 Connect Four AI

Neural network AI for Connect Four (7x6 board).

Model

Input: 42 neurons, Hidden: 64+64+32 (tanh), Output: 1 (linear).

Training: Adam optimizer, MSE loss, Batch=10, Epochs=30, Val=20%, Test=10%, Early Stop=5.

Files

- `model.py` - NN model with victory check
- `main.ipynb` - Jupyter notebook
- `test-case-parser.py` - Test parser
- `test/` - Datasets

5 Neural Networks

- `neural_network/` - Main NN with jsonExtractor
- `neural_network_jr/` - Junior version

6 Technologies

C++: Disservin, nlohmann/json, STL. Python: TensorFlow, NumPy, scikit-learn, Jupyter.

7 Installation

Compile: `g++ implementations/*.cpp traditional_evaluation/*.cpp -I. -o chess_engine`

Run: `./chess_engine` (UCI notation: e2e4, etc.)

Train Connect Four: `cd connect_four \` and run jupyter notebook `main.ipynb` or `python model.py`

8 License

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9 Status

Completed: Chess engine, Alpha-Beta, evaluation system, Connect Four AI, victory check.

In Progress: Advanced NN for chess, move explorer, multi-threading, opening book.

10 Contributing

Llorenc Perez - perezllorenc@gmail.com - <https://github.com/LLPB-pixel/ajedrezUPV>

11 Acknowledgments

UPV, Disservin, nlohmann, TensorFlow team, Lichess.

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<https://github.com/LLPB-pixel/ajedrezUPV>